

The Time lines

384-322 BC Aristotle

1564-1642 Galileo : dialogue 1632

1643-1727 Newton principia

1736-1813 Lagrange

1805-1865 Hamilton

$$\eta = 1 - \frac{T_L}{T_H}$$

thermodynamics

Coulomb

1824 S Carnot

Faraday

1840s Joule

1831-1879 Maxwell : 1865 light & EM.

1850 Clausius

1850 Kelvin

1844-1906 Boltzmann

1839-1903 Gibbs

1905

Einstein

1915

Planck

Bohr

Heisenberg

Schrödinger

} 1925-1926

Born

Dirac

elementary particle theory
 gauge quantum fields
 string theory

"Thermodynamics & an introduction to Thermostatistics"

2nd Ed. H. B. Callen

Date

Atomic time scale / length scale 10^{-14} s , $1 \text{ \AA}$

macroscopic scale $\frac{1}{100} \text{ s}$, 10^{-6} m

vibrational frequencies of hydrogen - Carbon system is 3000 cm^{-1}

$$\frac{\nu}{c} = 3000 \text{ cm}^{-1} \rightarrow \nu = 3 \times 10^8 \text{ m} \times 3000 \frac{1}{\frac{1}{100} \text{ m}} \approx 10^{14} \text{ s}^{-1} \text{ (infrared)}$$

quantum picture

electrons are even faster

consider H atom

energy
size

$$E = 13 \text{ eV}$$

$$L = 0.5 \text{ \AA}$$

$$\hbar \nu \approx 10 \text{ eV}$$

$$\nu = \frac{10 \text{ eV}}{\hbar} = \frac{10 \text{ eV}}{6 \times 10^{-34} \text{ J.s}}$$

$$= \frac{10 \times 1.6 \times 10^{-19} \text{ J}}{6 \times 10^{-34} \text{ J.s}} = 10^{15} \text{ Hz} \text{ also in the same range ?}$$

classical picture

However

$$E = m \frac{L^2}{T^2} \rightarrow T = \sqrt{\frac{m L^2}{E}}$$

$$= \sqrt{\frac{10^{-3} \text{ amu} \cdot 0.25 \text{ \AA}^2}{13 \text{ eV}}}$$

$$= 4.3 \times 10^{-13} \sqrt{\frac{\text{amu} \text{ \AA}^2}{\text{eV}}} = 10^{-2} \cdot 10^{-14} \text{ s}$$

electrons should be 2 to 3 orders faster!

Thermodynamics is for macroscopic system with many particles (10^{23})
macroscopic measurements are slow (10^{-2} s) and coarse (10^{-3} m).

Conservation of Energy

Date

1693	Leibniz	$\frac{1}{2}mv^2 + mgh$	mechanical point mass
		$\frac{Q_1 Q_2}{r}$	electric (as well as field eng)
1905	Einstein	$\frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$	rest mass energy

1930 Enrico Fermi proposed neutrino for energy conservation

conservation of $E \rightleftharpoons$ time translational invariance
are the same today, eons ago, and remote future.
fundamental laws of physics

1798 Rumford } heat energy
1840's James Joule }

Callen's postulates on thermal dynamics 0-th law

Postulate I. There exist particular states (called equilibrium states) of simple systems that, macroscopically, are characterized completely by the internal energy U , the volume V , and mole number N_i of chemical components.

Existence of Equilibrium: static, independent of pass history
completely determined by extensive parameters

quiescence: not change temporarily.
Not enough

U, V, N . the δ parameters characterize the system is small.
small set of

ΔU of two states

Energy difference can be measured by purely mechanical means.

Microscopic state need a large number of variables.

$(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, \vec{p}_1, \vec{p}_2, \dots, \vec{p}_N)$

first law

Quantitative definition of heat

Date

$$Q = \Delta U - W$$

$\uparrow$ heat flux to a system $\nwarrow$ mechanical energy (work) to the system

$\Delta U = U_2 - U_1$ internal energy increase

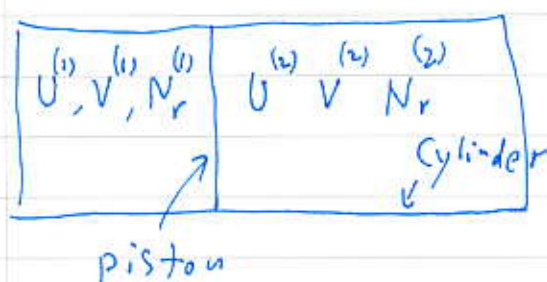
ΔU can be measured in some adiabatic process.

const. N .

For simple system $\delta W = -P dV$ work done to the system is positive.
 in quasi-static process (infinite slow) so that essentially the system is always in equilibrium.

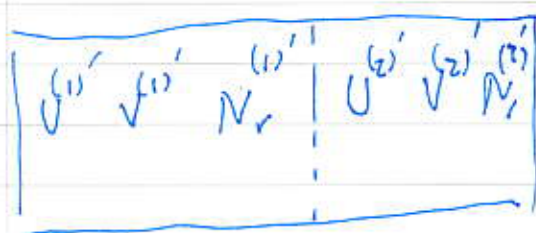
units for heat 1 calorie = 4.184 Joule

The single, all-encompassing problem of thermodynamics is the determination of the equilibrium state that eventually results after the removal of internal constraints in a closed, composite system.



initial two isolated systems
a composite system.

piston is rigid, adiabatic,
no holes.



what are the new states
when internal constraints
(the piston) is removed?
i.e. piston can move
or able to transfer heat
or transfer molecules?

The Entropy Maximum Postulates

2nd law

Date

Postulate II. There exists a function C called the

entropy S) of the extensive parameters of any composite system, defined for all equilibrium states and having the following property: The values assumed by the extensive parameters in the absence of an internal constraint are those that maximize the entropy over the manifold of constrained equilibrium states.

Max S meaning full only for composite system

$S(U, V, N_r \dots)$ is known as a fundamental relation.

If the fundamental relation of a particular system is known all conceivable thermodynamic information about the system is ascertainable from it.

[ascertainable]

Postulate III. The entropy of a composite system is additive over the constituent subsystems. The entropy is continuous & differentiable and is a monotonically increasing function of the energy.



total

$$S = S^{(1)} + S^{(2)} \quad \text{additivity}$$

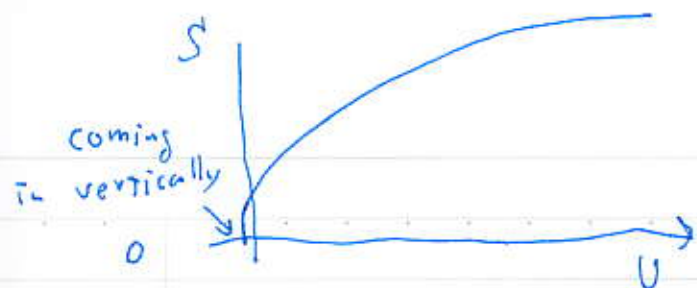
apply to single homogeneous system

$$S(\lambda U, \lambda V, \lambda N_r) = \lambda S(U, V, N_r)$$

monotonicity differentiability

$$\left(\frac{\partial S}{\partial U} \right)_{V, N_r} > 0$$

$$\left(\frac{1}{T} = \frac{\partial S}{\partial U} \right)_{\text{def}}$$



continuity
differentiability
monotonicity

Date

$S = S(U, V, N_r, \dots)$ can be solved to get $U = U(S, V, N_r, \dots)$

also
in fundamental
relations

Postulate IV. The entropy of any system vanishes in the state for which

$$\left(\frac{\partial U}{\partial S}\right)_{V, N_r} = 0 \quad (\text{that is, at the zero of temperature})$$

(3rd law
of thermodynamics)

differentials

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V, N_1, N_2, \dots, N_r} dS + \left(\frac{\partial U}{\partial V}\right)_{S, N_1, N_2, \dots, N_r} dV + \sum_{j=1}^r \left(\frac{\partial U}{\partial N_j}\right)_{S, V, \dots, N_r} dN_j$$

$$= T dS - P dV + \sum_{j=1}^r \mu_j dN_j$$

$$\left(\frac{\partial U}{\partial S}\right)_{V, N} \equiv T$$

temperature

$$\left(\frac{\partial U}{\partial V}\right)_{S, N} \equiv -P$$

pressure

$$\left(\frac{\partial U}{\partial N_j}\right)_{S, V} \equiv \mu_j$$

chemical potential
electro-

these three
quantities as
a function of
 S, V, N_j is also
known
as
equations
of state.

$$dW = -P dV \quad dQ = T dS \rightarrow dS = \frac{dQ}{T}$$

for quasi-static process. N fixed

"A quasi-static flux of heat into a system is associated with an increase of entropy of that system."

T, P, μ_j are intensive parameters in the sense, eg.

$$T(\lambda S, \lambda V, \lambda N_r) = T(S, V, N_r)$$

Definition 4 Temperature

$$U = U(S, V, N_1, N_2, \dots, N_r)$$

$$dU = \left(\frac{\partial U}{\partial S} \right)_{V, N_1, \dots, N_r} dS + \left(\frac{\partial U}{\partial V} \right)_{S, N_1, \dots, N_r} dV + \sum_j \left(\frac{\partial U}{\partial N_j} \right)_{S, V, N_r} dN_j$$

$$= T dS - P dV + \sum_j \mu_j dN_j$$

$$\left(\frac{\partial U}{\partial S} \right)_{V, N_1, \dots, N_r} \equiv T \quad \text{temperature}$$

$$\left(\frac{\partial S}{\partial U} \right)_{U, N} = \frac{1}{T}$$

$$\left(\frac{\partial U}{\partial V} \right)_{S, N} = -P$$

are these two
consistent?
yes!

However, we can also use

$$S = S(U, V, N_1, \dots, N_r)$$

$$dS = \left(\frac{\partial S}{\partial U} \right)_{V, N_1, \dots, N_r} dU + \left(\frac{\partial S}{\partial V} \right)_{U, N_1, \dots, N_r} dV + \sum_j \left(\frac{\partial S}{\partial N_j} \right)_{U, V, N_r} dN_j$$

(not helpful to write this)

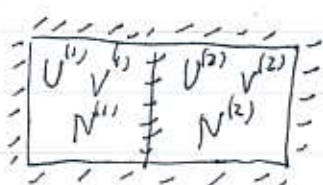
$$\left(\frac{\partial U}{\partial S} \right)_{V, N_1, \dots, N_r} \left(\frac{\partial S}{\partial U} \right)_{U, N_1, \dots, N_r} = 1$$

so we also have $\left(\frac{\partial S}{\partial U} \right)_{V, N_1, \dots, N_r} \equiv \frac{1}{T}$

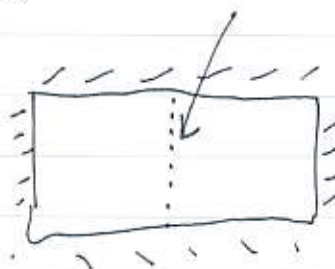
consider $0 = dU = \left(\frac{\partial U}{\partial S} \right) dS + \left(\frac{\partial U}{\partial V} \right) dV$
 $= T dS - P dV = 0$
 so $\frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{U, N}$

Thermal Equilibrium — temperature

diathermal wall



adiabatic walls



closed system

both, U, N, V

cannot change

two separate independent
simple systems with rigid, impermeable wall.

$V^{(1)}, V^{(2)}$ do not change
 $N^{(1)}, N^{(2)}$ because of the external constraints
 (rigid wall & piston)
 impermeable

So the only variables that can change due to the removal of internal constraints (replace adiabatic wall by diathermal wall) is $U^{(1)} + U^{(2)}$ with the overall constraints

$$U^{(1)} + U^{(2)} = \text{const.} = U$$

$$S = S^{(1)} + S^{(2)} = S^{(1)}(U^{(1)}, V^{(1)}, N^{(1)}) + S^{(2)}(U^{(2)}, V^{(2)}, N^{(2)})$$

$$dV^{(1)} = dV^{(2)} = 0 \quad \text{particle numbers}$$

$$dN^{(1)} = dN^{(2)} = 0 \quad \text{volume is fixed}$$

$$dS = \frac{\partial S^{(1)}}{\partial U^{(1)}} dU^{(1)} + \frac{\partial S^{(2)}}{\partial U^{(2)}} dU^{(2)}$$

$$dU^{(1)} + dU^{(2)} = 0$$

$$= \left(\frac{\partial S^{(1)}}{\partial U^{(1)}} \right)_{V^{(1)}, N^{(1)}} - \left(\frac{\partial S^{(2)}}{\partial U^{(2)}} \right)_{V^{(2)}, N^{(2)}} dU^{(1)} = 0 \quad \text{at the new equilibrium state.}$$

$$= \left(\frac{1}{T^{(1)}} - \frac{1}{T^{(2)}} \right) dU^{(1)} = 0 \rightarrow \frac{1}{T^{(1)}} = \frac{1}{T^{(2)}} \rightarrow T^{(1)} = T^{(2)}$$

* condition of thermal equilibrium is that temperature must be the same.

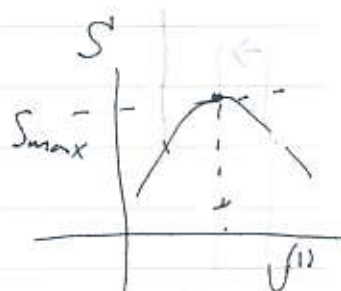
Intuitive concept of temperature (heat flow from low T to high T)

Let assume $T^{(1)} > T^{(2)}$ initially $(T^{(1)} \approx T^{(2)})$

$$\Delta S = S_{\text{final eq}} - S_{\text{initial eq}}$$

$$\Delta S = \left(\frac{1}{T^{(1)}} - \frac{1}{T^{(2)}} \right) \Delta U^{(1)} > 0 \rightarrow \Delta U^{(1)} < 0$$

The hotter one has initial energy decrease!



Units for temperature

$k_B T$ is joule

the $\frac{S}{k_B}$ is dimensionless

$$dU = TdS - PdV + \mu dN$$

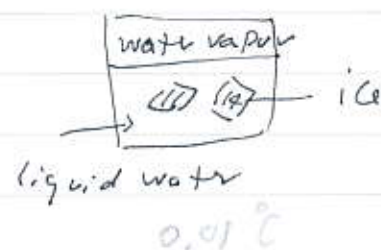
energy $\uparrow$ $\uparrow$ Kelvin $[S] = \text{joule/Kelvin}$
 IS units is joule symbol K units for energy

The Kelvin scale $T=0$ is minimal possible temperature fixed by nature

$T_+ = 273.16$ at the triple pts of water

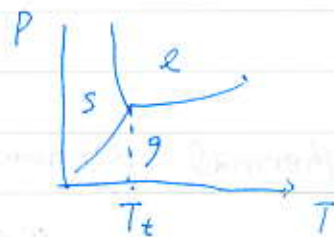
Celsius scale

$$T(^{\circ}C) = T(K) - 273.15$$

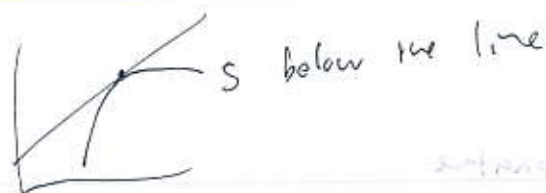


Fahrenheit scale

$$T(^{\circ}F) = \frac{9}{5} T(^{\circ}C) + 32$$



An overall T character for equilibria of composite systems



②

concavity implies $C_V > 0$

proof: $\left(\frac{\partial^2 S}{\partial U^2} \right)_{V,N} \leq 0$

(local stability condition)

$$\left(\frac{\partial^2 S}{\partial U^2} \right)_{V,N} = \frac{\partial}{\partial U} \left(\frac{\partial S}{\partial U} \right)_V = \left[\frac{\partial}{\partial U} \left(\frac{1}{T} \right) \right]_V = - \frac{1}{T^2} \left(\frac{\partial T}{\partial U} \right)_V$$

$$= - \frac{1}{T^2} \frac{1}{C_V} \leq 0 \quad T \geq 0$$

$$\rightarrow C_V \geq 0$$

$$\delta Q = T dS$$

thermal dynamic functions (second derivatives)

heat capacity

$$C_V \equiv \left(\frac{\delta Q}{dT} \right)_{V,N} = T \left(\frac{\partial S}{\partial T} \right)_{V,N} = \left(\frac{\partial U}{\partial T} \right)_{V,N} \quad \text{enthalpy}$$

$$C_P \equiv \left(\frac{\delta Q}{dT} \right)_{P,N} = T \left(\frac{\partial S}{\partial T} \right)_{P,N} = \left(\frac{\partial H}{\partial T} \right)_{P,N} \quad \begin{array}{l} H = U + PV \\ \text{Helmholtz} \\ \text{free energy} \end{array}$$

coefficient of thermal expansion

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

isothermal compressibility

$$K_T = - \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

$$C_P = C_V + \frac{TV\alpha^2}{K_T}$$

Some Formal relationships

- (A) Euler equation U is extensive for single species of single systems

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

$\frac{\partial}{\partial \lambda}$ on both side, set $\lambda = 1$

$$\frac{\partial U(\lambda S, \lambda V, \lambda N)}{\partial (\lambda S)} S + \frac{\partial U(\lambda S, \lambda V, \lambda N)}{\partial (\lambda V)} V + \frac{\partial U(\lambda S, \lambda V, \lambda N)}{\partial (\lambda N)} N = U(S, V, N)$$

(1) $TS - PV + \mu N = U$

- (B) Gibbs-Duhem relation (the intensive parameters are all independent)

differentiate (1)

$$dU = TdS + SdT - PdV - VdP + \mu dN + Nd\mu$$

since $dU = TdS - PdV + \mu dN$

we find $0 = SdT - VdP + Nd\mu$

concavity of $S(U)$ see ^{up note} pge 101

- (C) Maxwell's relation $U(S, V, N)$

→ Mixed derivatives are equal $\frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}$

$$\rightarrow -\left(\frac{\partial P}{\partial S}\right)_{V, N} = \left(\frac{\partial T}{\partial V}\right)_{S, N}$$

Dynamics

Date

statistical mechanics $\rightarrow$ interpret/principles thermodynamics from atomic point of view.

classical dynamics

Newton's law $F = ma$

"principle of least action"

Lagrange formulation $\delta \int \mathcal{L} dt = 0 \rightarrow$ Euler-Lagrange Eq. $\dot{q} = \frac{dq}{dt}$

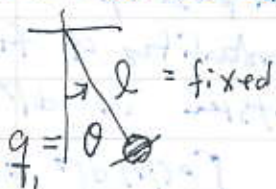
where $\mathcal{L} = K - V$

Hamilton's formulation

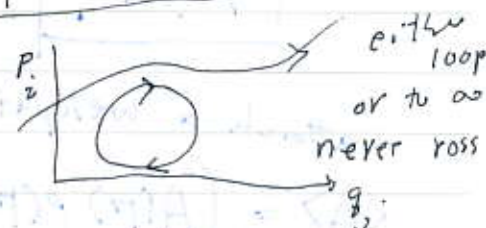
$$\mathcal{H}(p, q) = \sum_{i=1}^s p_i \dot{q}_i - \mathcal{L} \quad p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

beautify 4 Lagrange/Hamilton's formulati $q_1, q_2 \dots q_s$ can be any variables uniquely specify a system
the equations are the same in any coordinate system

pendulum



phase space (p, q)



①

Hamilton's Eq.

$$\dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}$$

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}$$

$$A(q_1, q_2 \dots p_1 \dots p_s)$$

Poisson brackets

$$(A, B) = \sum_i \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial B}{\partial q_i} \frac{\partial A}{\partial p_i} \right)$$

No explicit t dependence

$$(p_j, \mathcal{H}) = \dot{p}_j$$

also

$\Rightarrow$

$$(A, \mathcal{H}) = \dot{A}$$

equation valid for any time

$$(q_j, \mathcal{H}) = \dot{q}_j$$

Solution $\uparrow$ any arbitrary function of p, q independent

$$A_t = e^{t[H, A]}$$

No explicit t

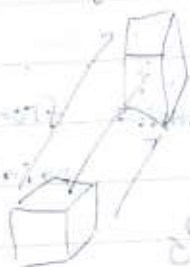
$$(\mathcal{H}, \mathcal{H}) = \ddot{\mathcal{H}} = 0 \quad \text{Energy is conserved}$$

$$A_t = A(q_t, p_t) = A(q, p, t) \quad q = q_{t=0} \quad p = p_{t=0} \quad A = A_{t=0} = A(p, q)$$

$$= A(q, p, 0) + \dot{A}_{t=0} t + \frac{\ddot{A}_{t=0}}{2} t^2 + \dots = A + (A, \mathcal{H}) t + \frac{(A, \mathcal{H}), \mathcal{H}}{2} t^2 + \dots$$

total differential
 $t + \Delta t$ later

$$\frac{DP}{Dt} = 0 = \frac{\partial P}{\partial t} + \sum_i \dot{q}_i \frac{\partial P}{\partial q_i} + \sum_i \dot{p}_i \frac{\partial P}{\partial p_i}$$



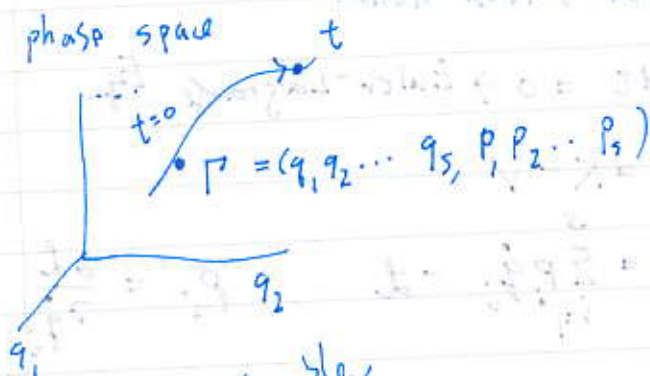
$$= \frac{\partial P}{\partial t} + \sum_i \left[\frac{\partial H}{\partial p_i} \frac{\partial P}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial P}{\partial p_i} \right]$$

$$= \frac{\partial P}{\partial t} + [P, H] = 0$$

volume preserving

(2)

phase space



S : degrees of freedom

ensemble



distribution of points

$P(P)$

$P(P) dP$ = # of systems in phase space volume dP or probability of finding system in dP

attach a weight to each system

$$\int P(P) dP = 1$$

$$P(P) \geq 0$$

$$\langle A \rangle = \int A(P) P(P) dP$$

ensemble average at particular time

Dynamics: The weight is fixed to a given system



system 1
 appears at
 point P_1



system 2
 appears at
 point P_2

different time A is different

See fixed t

$$\langle A \rangle_t = \int A(P, t) P(P) dP = \int dP P(P) e^{t(P, H)} A(P)$$

change of variable $P \rightarrow P_t$

$$= \int A(P_t) P(P_t) dP_t$$

$$= \int A(P_t) P(P_t) \frac{\partial P}{\partial P_t} dP_t$$

via Jacobian $P_t = e^{t(P, H)} P$

Symplectic

property of Hamiltonian system

differential

2-form

the dynamics preserved

$$dp \wedge dq$$

$$= -dq \wedge dp$$

$$\omega^2 = \sum_i dp_i \wedge dq_i$$

$$(\omega^2)^k$$

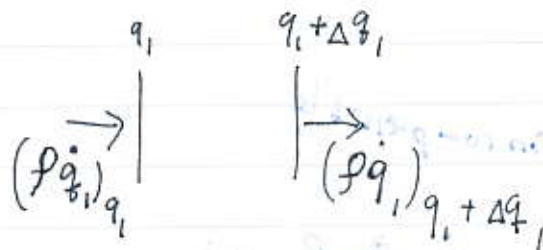
$$(\omega^2)^k = \underbrace{\omega^2 \wedge \omega^2 \wedge \dots \wedge \omega^2}_k$$

$$(\omega^2)^5 = dp_1 \wedge dp_2 \wedge \dots \wedge dp_5 \wedge dq_1 \wedge dq_2 \wedge \dots \wedge dq_5 = \text{phase space volume}$$

Liouville's theorem

$$\rho(q_1, q_2, \dots, p_1, \dots, p_s)$$

Gibbs picture of ensemble probability of finding the system



$$F_1 = \text{flux in } p \dot{q}_1 \Delta q_2 \Delta q_3 \dots \Delta p_s$$

surface perpendicular to axis q_1

$$F_2 = \text{flux out } (p \dot{q}_1)_{q_1 + \Delta q_1} \Delta q_2 \Delta q_3 \dots \Delta p_s$$

$$\sigma = (p \dot{q}_1 + \frac{\partial p \dot{q}_1}{\partial q_1} \Delta q_1)$$

$$\frac{\partial \rho}{\partial t} \Delta q_1 \Delta q_2 \dots \Delta p_s = F_1 - F_2 = \sum_j \left(\frac{\partial p \dot{q}_j}{\partial q_j} + \frac{\partial p \dot{p}_j}{\partial p_j} \right) \Delta q_1 \Delta q_2 \dots$$

$$\frac{\partial \rho}{\partial t} = \sum_j - \dot{q}_j \frac{\partial \rho}{\partial q_j} - \dot{p}_j \frac{\partial \rho}{\partial p_j} + \rho \left(\frac{\partial \dot{q}_j}{\partial q_j} + \frac{\partial \dot{p}_j}{\partial p_j} \right)$$

Partial derivative

$$= \sum_j - \frac{\partial H}{\partial p_j} \frac{\partial \rho}{\partial q_j} + \frac{\partial H}{\partial q_j} \frac{\partial \rho}{\partial p_j} + \rho \left(\frac{\partial^2 H}{\partial q_j^2} + \frac{\partial^2 H}{\partial p_j^2} \right) = 0$$

$$= (H, \rho)$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho \dot{x}) = 0 \text{ generates to } 2N\text{-dim}$$

use on the mechanical analog

compare conservation Eq in elements

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

fluid dynamics incompressible fluid

not need to define

Balescu

Γ_t remains to Γ

$$\frac{\partial \Gamma}{\partial t} = 1$$

$$\langle A \rangle_t = \int A(\Gamma) \rho(e^{-t(\cdot, H)} \Gamma) d\Gamma$$

$$= \int A(\Gamma) e^{-t(\cdot, H)} \rho(\Gamma) d\Gamma = \int A(\Gamma) \rho(t) d\Gamma$$

$$\rho(t) = e^{-t(\cdot, H)} \rho(0)$$

$$\frac{\partial \rho}{\partial t} = -(\cdot, H) \rho = (H, \rho)$$

Liouville eq

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial p} \cdot (p \dot{q}) = 0 \quad \text{incompressible}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial q} \rho \frac{\partial}{\partial p} \cdot \dot{q} + \dot{q} \cdot \frac{\partial}{\partial p} \rho = 0$$

$$\frac{\partial \rho}{\partial t} + \sum \dot{q} \frac{\partial}{\partial q} \rho \frac{\partial}{\partial p} = 0 \Rightarrow \frac{D \rho}{D t} = 0$$

$$\frac{\partial \rho}{\partial t} + \sum \frac{\partial H}{\partial q} \frac{\partial \rho}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial \rho}{\partial q} = 0$$

$$\frac{\partial \rho}{\partial t} + (\rho, H) = 0 \rightarrow \frac{\partial \rho}{\partial t} = -(\rho, H)$$

Hilbert space

$$(\phi, \psi) = (\psi, \phi)^* = \langle \phi | \psi \rangle$$

$$(\phi, \phi) = \langle \phi | \phi \rangle = \|\phi\|^2 \geq 0 \Leftrightarrow \phi = 0$$

$$(\phi, c\psi) = c(\phi, \psi)$$

$$(\phi, \psi + \chi) = (\phi, \psi) + (\phi, \chi)$$

Hermitian

$$A = A^\dagger$$

$$(\phi, A\psi) = (A^\dagger \phi, \psi)$$

$$A\phi = \lambda\phi$$

λ is an eigenvalue of A and ϕ is an eigenvector

Quantum dynamics

State $\psi(q_1, q_2, \dots, q_s^t)$ wave function

$$p_j = \frac{\hbar}{i} \frac{\partial}{\partial q_j}$$

$$i\hbar \frac{\partial \psi}{\partial t} = H \psi$$

$$H\left(\frac{\hbar}{i} \frac{\partial}{\partial q_1}, \dots, q_1, q_2, \dots, q_s\right)$$

given $\psi(t=0)$ we can predict $\psi(t)$ for any t

$$\rho = |\psi\rangle\langle\psi|$$

$$A\rho \quad \rho^2 = \rho \quad \text{projector}$$

$$\langle A \rangle_{QM} = \langle \psi(t) | A | \psi(t) \rangle = \int dq_1 dq_2 \dots dq_s \psi^* A \psi = \text{Tr}(A |\psi\rangle\langle\psi|)$$

Heisenberg picture

$$= \langle \psi(0) | e^{\frac{iHt}{\hbar}} A e^{-\frac{iHt}{\hbar}} | \psi(0) \rangle = \langle \psi(0) | A(t) | \psi(0) \rangle$$

$$A(t) = e^{\frac{iHt}{\hbar}} A e^{-\frac{iHt}{\hbar}} \quad | \psi(0) \rangle = e^{\frac{iHt}{\hbar}} | \psi(t) \rangle$$

Heisenberg Eq. of motion

the Heisenberg picture wave function $\rightarrow$ the Schrödinger picture wave

$$\begin{aligned} \dot{A}(t) &= e^{\frac{iHt}{\hbar}} \left(\frac{iH}{\hbar} A \right) e^{-\frac{iHt}{\hbar}} - e^{\frac{iHt}{\hbar}} A \frac{iH}{\hbar} e^{-\frac{iHt}{\hbar}} \\ &= \frac{1}{i\hbar} (A(t)H - HA(t)) = \frac{1}{i\hbar} [A(t), H] \end{aligned}$$

$$= (A(t), H)$$

quantum poisson bracket

$$(A, B) = \frac{1}{i\hbar} (AB - BA)$$

$$\rightarrow \dot{A} = (A, H)$$

$$\frac{\partial^2}{\partial t^2} = (H, p) = \frac{\partial H}{\partial q} \frac{\partial p}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial p}{\partial q} = -\dot{p} \frac{\partial p}{\partial p} - \dot{q} \frac{\partial p}{\partial q}$$

$$\frac{Dp}{Dt} = \frac{\partial p}{\partial t} + \dot{q} \frac{\partial p}{\partial q} + \dot{p} \frac{\partial p}{\partial p} = 0$$

Books to read:

Balescu: chaps 1, 2, 4

K Huang: chapter 1, 6, 8

Plischke & Bergersen: chapter 1, 2

Toda Rube Saito: chapter 1, 2

Richard C Tolman: the principles of Stat Mech

$$J = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$$

canonical transform

$$0) \quad \dot{x} = J \frac{\partial H}{\partial x} \Rightarrow \Gamma = J \frac{\partial K}{\partial \Gamma} \quad K = H + \dots \quad \text{Date}$$

$$x = \begin{pmatrix} p \\ q \end{pmatrix} \quad \Gamma = \begin{pmatrix} P \\ Q \end{pmatrix} \quad p dq - H = P dQ - K + \frac{dF}{dt}$$

$$\Gamma = \Gamma(x, t)$$

$$1) \quad \omega^2 = \sum dp \wedge dq \quad \omega^2 = \Omega^2$$

$$\Omega^2 = \sum dQ \wedge dP$$

$$2) \quad M^T J M = J \quad (M)_{ij} = \frac{\partial \Gamma_i}{\partial x_j} \quad \det(M) = 1$$

$$3) \quad [Q_i, P_j]_x = \delta_{ij}$$

$$(Q_i, Q_j) = 0 \quad \rightarrow (A, B) = \left(\frac{\partial A}{\partial x} \right)^T J \frac{\partial B}{\partial x}$$

$$(P_i, P_j) = 0$$

$$4) \quad \Gamma = \Gamma(x) = e^{\alpha(x)} \quad \text{is a canonical transform}$$

set of all canonical transforms form a group

classical Poisson bracket

$$(A, B) = \sum_j \frac{\partial A}{\partial q_j} \frac{\partial B}{\partial p_j} - \frac{\partial B}{\partial q_j} \frac{\partial A}{\partial p_j}$$

↑
arbitrary functions of canonical variables p, q

Quantum version

$$(A, B) = \frac{1}{i\hbar} [A, B]$$

↑
operators in Hilbert space

common properties of Poisson bracket

① antisymmetric

$$(A, B) = -(B, A)$$

② linear

$$(A+B, C) = (A, C) + (B, C)$$

$$(\lambda A, C) = \lambda (A, C)$$

↑ constant

③ Jacob identity

$$((A, B), C) + ((B, C), A) + ((C, A), B) = 0$$

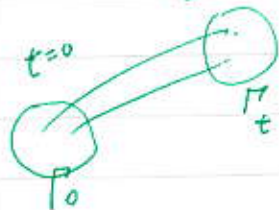
④ Leibniz rule

$$(AB, C) = A(B, C) + (A, C)B$$

C 's action is like a differential operator

$$(q_j, p_k) = \delta_{jk} \quad \text{true in both classical \& quantum mechanics}$$

1. should that the Jacobian of the transform $\Gamma_0 \rightarrow \Gamma_t$ for fixed t is 1.



$$D(t) = 1$$

compute $\dot{D}(t)$

$$D = \det \left(\frac{\partial x_j(t)}{\partial x_i(0)} \right)$$

$$X = (q \dots p \dots)$$

2. show

$$\frac{d\langle A \rangle}{dt} = \left\langle \frac{dA}{dt} \right\rangle$$

$$\langle A \rangle = \int A \rho d\Gamma$$

where we assume term term is zero!

Density Matrix

a pure state.

for a single quantum state $\rho = |\psi\rangle\langle\psi|$
 can be used instead of $|\psi\rangle$ to represent a state

Let $|\psi_i\rangle$ be arbitrary orthonormal states

$$\rho = \sum_i w_i |\psi_i\rangle\langle\psi_i|$$

← spectrum decomposition of ρ

Schrödinger
representant
object

$$\rho |\psi_i\rangle = w_i |\psi_i\rangle$$

← eigenvalues problem of ρ

$$w_i > 0, \quad \sum_i w_i = 1$$

positive
-definite
operator

$$\text{Tr}(\rho) = 1$$

statistical average

$$\langle A \rangle = \text{Tr}(\rho A) = \sum_i w_i \text{Tr}(|\psi_i\rangle\langle\psi_i| A) = \sum_i w_i \langle\psi_i| A |\psi_i\rangle$$

QM average

$$\dot{\rho} = \sum_i w_i \left(\frac{\partial |\psi_i\rangle}{\partial t} \langle\psi_i| + |\psi_i\rangle \frac{\partial \langle\psi_i|}{\partial t} \right)$$

$$= \sum_i w_i \left(\frac{1}{i\hbar} H |\psi_i\rangle \langle\psi_i| + |\psi_i\rangle \left(-\frac{1}{i\hbar} \langle\psi_i| H \right) \right)$$

$$= \frac{1}{i\hbar} (H\rho - \rho H) = (H, \rho)$$

← Quantum Liouville eq.
or von Neumann eq.

formal solution of $\dot{\rho} = (H, \rho) = -i\mathcal{L}\rho$

$$\rho(t) = e^{-i\mathcal{L}t} \rho(0) = e^{-\frac{iH}{\hbar}t} \rho(0) e^{\frac{iH}{\hbar}t}$$

$$= e^{t\mathcal{L}} \rho(0)$$

$$i\hbar \dot{\rho} = [H, \rho]$$

32 relation between classical & quantum descriptions

b $\rho(p, q)$ $\hat{\rho}$

Wigner's Distribution function

1D: $\rho_w(p, q) = \int_{-\infty}^{\infty} dr e^{-i p r / \hbar} \langle x | \hat{\rho} | x' \rangle$
 $q = \frac{x+x'}{2}$
 $r = x - x'$

then $\langle A \rangle = \text{Tr}(\hat{\rho} \hat{A})$

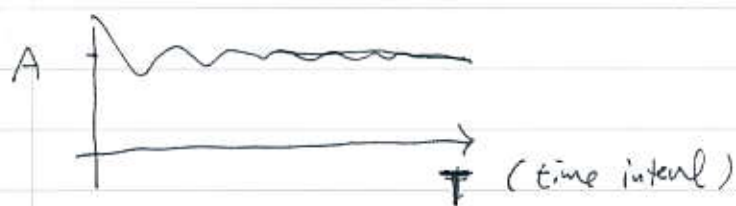
$= \int \frac{dp dq}{h} A_w(q, p) \rho_w(q, p)$

$\hbar \rightarrow 0$
 $\rho(p, q) = \frac{1}{(\Delta q \Delta p)} \left(\rho_w(q, p) \Delta q \Delta p + O(\hbar^2) \right)$
 $\Delta q \Delta p \gg \hbar$

	classical	quantum
state	a point in phase space $\Gamma = \begin{pmatrix} p_1 \\ p_2 \\ q_1 \\ q_2 \\ q_3 \end{pmatrix}$	wave funct $\psi \rightarrow \psi(q_1, q_2, \dots, q_r)$
observables	$A(\Gamma)$	hermitian operator $\hat{A}$
distribn	$\rho(\Gamma)$ Liouville eq	$\hat{\rho}$ von Neumann eq $\frac{\partial \rho}{\partial t} = (H, \rho)$
avg	$\int d\Gamma A \rho$	$\text{tr}(\hat{A} \hat{\rho})$
input Math object	$\{A, B\} = \sum \dots$ Poisson brackets	commutator $\frac{1}{\hbar} [\hat{A}, \hat{B}]$

macroscopic
classical observables

$$\bar{A} = \frac{1}{T} \int_0^T A(p_1^H, p_2^H, \dots, q_1^H, q_2^H, \dots, q_N^H) dt \quad \begin{matrix} T \text{ large} \\ \text{time ave} \end{matrix} \quad T \rightarrow \infty$$



when $T \rightarrow \infty$ the
system approaches eq.

$\bar{A}$ becomes indep of T

$\bar{A}$ can be computed on a computer by MD.
typical microscopic time scale 10^{-14} s

$$T \approx 10^6 \sim 10^9 \times 10^{-14} \text{ s} \approx \mu\text{s} = 10^{-6} \text{ s}$$

statistical mechanics method by-pass the explicit ave
process. replace it by ensemble ave

$$\langle A \rangle = \int \dots \int dp_1 dp_2 \dots dp_N dq_1 dq_2 \dots dq_N, \rho(p_1, p_2, \dots, q_1, q_2, \dots) A(p, q)$$

$$\bar{A} \stackrel{?}{=} \langle A \rangle$$

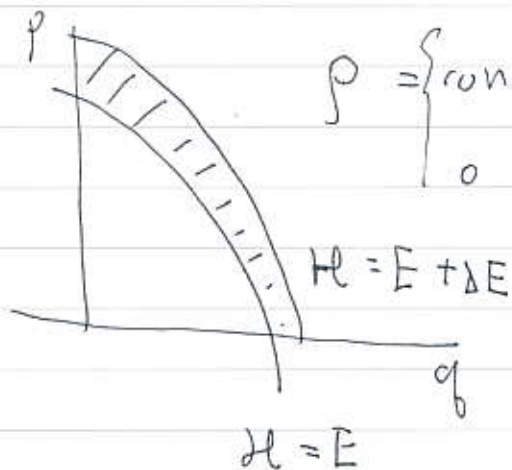
$$\int \dots \int dp \dots dq \rho(\dots) = 1$$

time ave equal ensemble
ave

Ergodic hypothesis:

Yes $\bar{A} = \langle A \rangle$

microcanonical ensemble



$$\rho = \begin{cases} \text{const} \\ 0 \end{cases}$$

$$E < \mathcal{H} < E + \delta E$$

$$\text{otherwise } \rho \propto \delta(\mathcal{H} - E)$$

$$S(E, V, N) \propto \log(\text{phase space volume})$$

$$\frac{\partial \rho}{\partial t} = 0 \quad (H, \rho)$$

Liouville eq.

the constant can be found by
normalization

Boltzmann's Principle

→ k_B due 1st by Planck gave it a
2% accurate value in study of
black body radiation
(if we use only T , to measure
temperature then $k_B = 1$)

Date

$$S = k_B \ln \Omega$$

↑ natural log (base e)

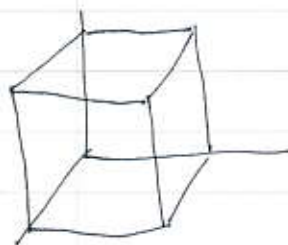
Boltzmann's constant $k_B = 1.38065 \times 10^{-23} \text{ J/K}$ *microstate are equally probable. i.e. microcanonical.*
(due to Planck 1900)

quantum

when a system is in a unique ground state $\Omega = 1, S = 0$
This is the 3rd law of thermodynamics

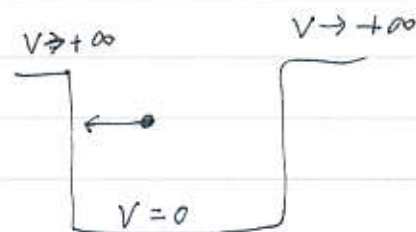
Example.

classical Ideal gas



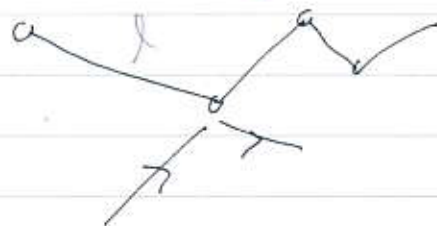
$$\mathcal{H}(p_1, p_2, \dots, q_1, q_2, \dots) = \sum_{j=1}^{3N} \frac{p_j^2}{2m} + V(q_1, \dots, q_{3N})$$

effect of walls + molecule interactions



$$-\infty < p_j < +\infty$$

$$0 \leq q_i < L, \quad \text{volume } V = L^3$$



dilute
mean-free path of gas

$$l \propto \frac{1}{\frac{N}{V} \cdot \sigma^2}$$

air at room T . $\sigma \approx 1 \text{ \AA}$ $\frac{N}{V} \approx 10^{25} \frac{1}{\text{m}^3}$

$$l \propto 10^{-8} \text{ m} \approx 100 \sigma$$

tutorial 1 Sat 20 Jan 07

Date

2.6-4:

$$\frac{1}{T^{(1)}} = \frac{3}{2} R N^{(1)} / U^{(1)} \quad \leftarrow \text{mole number}$$

$$\frac{1}{T^{(2)}} = \frac{5}{2} R N^{(2)} / U^{(2)} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{equation of state}$$

$$U^{(1)} = \frac{3}{2} N^{(1)} R T$$

Eq. of states

$$\frac{1}{T} = \frac{\partial S(U, V, N)}{\partial U} \quad |_{V, N}$$

$$P = - \frac{\partial U}{\partial V} |_{S, N}$$

$$\mu = \frac{\partial U}{\partial N} |_{S, V}$$

$$R = 8.314 \text{ J/(mole} \cdot \text{K)}$$



$$N^{(1)} = 2$$

$$T^{(1)} = 250 \text{ K}$$

$$N^{(2)} = 3$$

$$T^{(2)} = 350 \text{ K}$$

diathermal wall

compute the entropy $\frac{1}{T^{(1)}} = \frac{3}{2} R \frac{N^{(1)}}{U^{(1)}} = \frac{\partial S^{(1)}(U^{(1)}, V^{(1)}, N^{(1)})}{\partial U^{(1)}} \quad |_{V^{(1)}, N^{(1)}}$

Fixed volume and N , in the problem

$$S^{(1)} = \int \frac{3}{2} R \frac{N^{(1)}}{U^{(1)}} dU^{(1)} = \frac{3}{2} R N^{(1)} \ln U^{(1)} / U_0^{(1)} + \text{const (N, V)}$$

Similarly $S^{(2)} = \frac{5}{2} R N^{(2)} \ln U^{(2)} / U_0^{(2)}$

$$S = S^{(1)} + S^{(2)} = \frac{3}{2} R N^{(1)} \ln U^{(1)} / U_0^{(1)} + \frac{5}{2} R N^{(2)} \ln U^{(2)} / U_0^{(2)}$$

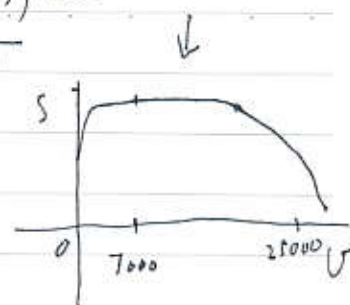
$$U = U^{(1)} + U^{(2)} = \text{const} = \frac{3}{2} N^{(1)} R T^{(1)} + \frac{5}{2} N^{(2)} R T^{(2)}$$

$$= \left(\frac{3}{2} \cdot 2 \cdot 250 + \frac{5}{2} \cdot 3 \cdot 350 \right) 8.314 \text{ J/(mole} \cdot \text{K)}$$

$$= 3 \times (250 + \frac{5}{2} \cdot 350) \times 8.314 = 28059.75 \text{ J}$$

S is concave

$$S = \frac{3}{2} R N^{(1)} \ln \frac{U^{(1)}}{U_0^{(1)}} + \frac{5}{2} R N^{(2)} \ln \frac{U - U^{(1)}}{U_0^{(2)}} \quad \text{Date}$$



find maximum of S

$$\frac{\partial S}{\partial U^{(1)}} = 0 = \frac{3}{2} R N^{(1)} \frac{1}{U^{(1)}} + \frac{5}{2} R N^{(2)} \frac{(-1)}{U - U^{(1)}}$$

$$\frac{3 N^{(1)}}{U^{(1)}} = \frac{5 N^{(2)}}{U - U^{(1)}} \rightarrow \frac{6}{U^{(1)}} = \frac{15}{U - U^{(1)}}$$

$$2(U - U^{(1)}) = 5 U^{(1)} \rightarrow 2U = 7 U^{(1)}$$

$$U_{eq}^{(1)} = \frac{2}{7} U = \frac{2}{7} \times 28059.75 \text{ J} = 8017.07 \text{ J}$$

$$U_{eq}^{(2)} = U - U^{(1)} = \frac{5}{7} U = 5726.48 \text{ J} = 20042.68 \text{ J}$$

$$T_{eq} = T_{eq}^{(1)} = T_{eq}^{(2)} = \frac{U_{eq}^{(1)}}{\frac{3}{2} R N^{(1)}} = \frac{8017.07 \text{ J}}{\frac{3}{2} \cdot 8.314 \text{ J (mol K)} \cdot 2 \text{ mol}} = 321.43 \text{ K}$$

$$T^{(1)} < T_{eq} < T^{(2)}$$

$$\begin{cases} U^{(1)} = \frac{3}{2} N^{(1)} R T^{(1)} \\ U^{(2)} = \frac{5}{2} N^{(2)} R T^{(2)} \end{cases} \quad \begin{cases} U^{(1)} + U^{(2)} = U \text{ fixed} \\ \frac{3}{2} \frac{U^{(1)}}{N^{(1)}} = \frac{5}{2} \frac{U^{(2)}}{N^{(2)}} \quad T = T_2 \end{cases}$$

$$\begin{aligned} U^{(1)} + \frac{5}{3} U^{(1)} &= U \\ \frac{8}{3} U^{(1)} &= U \\ U^{(1)} &= \frac{3}{8} U \\ U^{(2)} &= \frac{5}{8} U \end{aligned}$$

sec 2-7

$N^{(1)} = \text{fixed}$	$N^{(2)} = \text{fixed}$
$U^{(1)}$	$U^{(2)}$
$V^{(1)}$	$V^{(2)}$
	$\rightarrow$

Date

movable
diathermal
wall

$$U^{(1)} + U^{(2)} = U = \text{const}$$

$$V^{(1)} + V^{(2)} = V = \text{const}$$

$$S = S^{(1)} + S^{(2)}$$

when in equilibrium, there is no transfer of heat $T dS = dU = 0$

$$0 = dS = \left. \frac{\partial S^{(1)}}{\partial U^{(1)}} \right|_{V^{(1)}, N^{(1)}} dU^{(1)} + \left. \frac{\partial S^{(1)}}{\partial V^{(1)}} \right|_{U^{(1)}, N^{(1)}} dV^{(1)}$$

$$dU^{(1)} + dU^{(2)} = 0$$

$$dV^{(1)} + dV^{(2)} = 0$$

$$+ \left. \frac{\partial S^{(2)}}{\partial U^{(2)}} \right|_{V^{(2)}, N^{(2)}} dU^{(2)} + \left. \frac{\partial S^{(2)}}{\partial V^{(2)}} \right|_{U^{(2)}, N^{(2)}} dV^{(2)}$$

$dU^{(1)}$ or small
change
in each

$$= \left(\frac{1}{T^{(1)}} - \frac{1}{T^{(2)}} \right) dU^{(1)} + \left(\frac{P^{(1)}}{T^{(1)}} - \frac{P^{(2)}}{T^{(2)}} \right) dV^{(1)} = 0$$

$dS = 0$ for arbitrary $dU^{(1)}$ and $dV^{(1)}$

$$\rightarrow \frac{1}{T^{(1)}} = \frac{1}{T^{(2)}} \quad , \quad \frac{P^{(1)}}{T^{(1)}} = \frac{P^{(2)}}{T^{(2)}} \quad \rightarrow \quad P^{(1)} = P^{(2)}$$

$$dU = T ds - p dv + \mu dN$$

$$= \left(\frac{\partial U}{\partial S} \right)_{V, N} dS + \left(\frac{\partial U}{\partial V} \right)_{S, N} dV + \left(\frac{\partial U}{\partial N} \right)_{S, V} dN$$

$$\rightarrow dS = \frac{1}{T} dU + \frac{p}{T} dV - \frac{\mu}{T} dN \rightarrow \left(\frac{\partial S}{\partial U} \right)_{V, N} = \frac{1}{T}$$

$$\left(\frac{\partial S}{\partial V} \right)_{U, N} = \frac{p}{T}$$

$$\left(\frac{\partial S}{\partial N} \right)_{U, V} = -\frac{\mu}{T}$$

Ex 3-4 pgs 66-68

Date

$$dU = TdS - PdV + \mu dN$$

 $\frac{dU}{d \cdot}$

$$dS = \frac{1}{T} dU + \frac{P}{T} dV - \frac{\mu}{T} dN$$

 $S(U, V, N)$
let $dN = 0$

$$= \frac{CNK}{U} dU + \frac{NKT}{V} dV + 0$$

equations of state

$$\begin{cases} PV = NKT \\ U = CNKT \end{cases}$$

$$\int_{S_0}^S dS = \int_{U_0}^U \frac{CNK}{U} dU + \int_{V_0}^V \frac{NKT}{V} dV$$

$$S = S_0 +$$

$$S(U, V, N) = \underbrace{S_0(U_0, V_0, N)}_{\text{unknown constant}} + CNK \ln \frac{U}{U_0} + NK \ln \frac{V}{V_0}$$

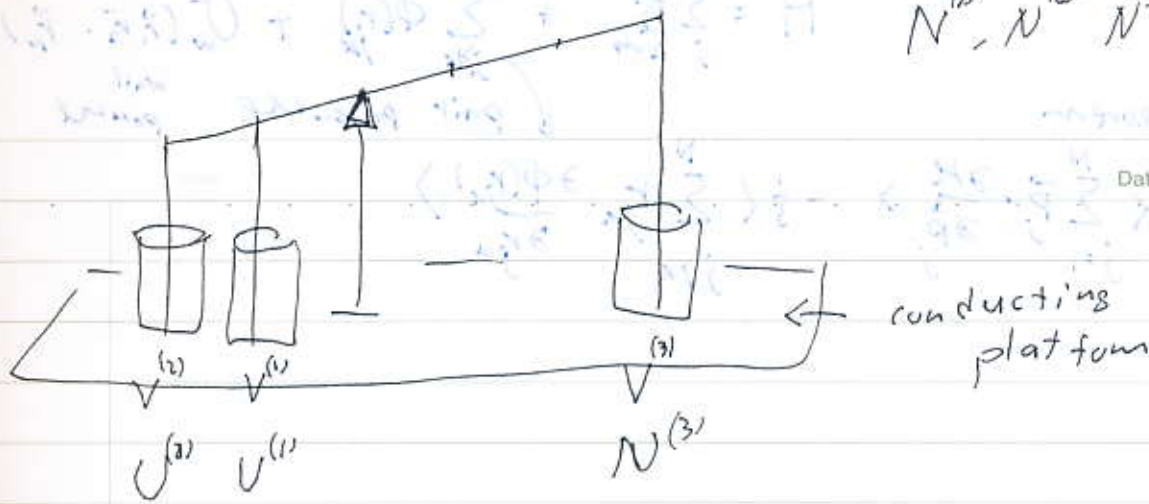
problem when $U \rightarrow 0$
or $V \rightarrow 0$

discuss equation

how to determine

$$S \text{ from } dS = f(U, V) dU + g(U, V) dV \quad ?$$

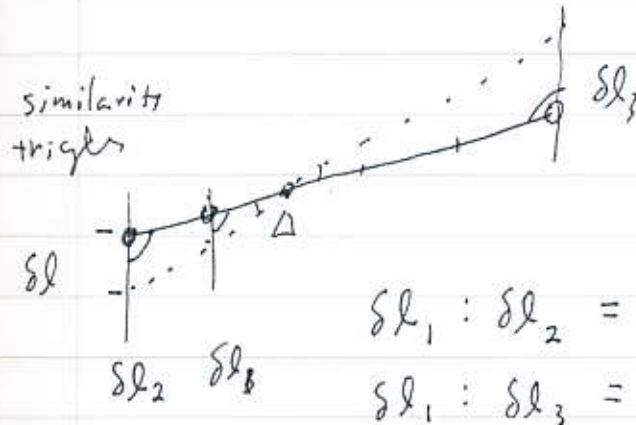
$$N^{(1)}, N^{(2)}, N^{(3)} = \text{fixed}$$



total system is isolated

$$U^{(1)} + U^{(2)} + U^{(3)} = U = \text{const}$$

$$S^{(1)} + S^{(2)} + S^{(3)} = S \uparrow \text{max}$$



$$\begin{aligned} \rightarrow 2 \delta V^{(1)} &= \delta V^{(2)} \\ \delta V^{(3)} &= -3 \delta V^{(1)} \end{aligned}$$

$$\delta l_1 : \delta l_2 = 1 : 2$$

$$\delta l_1 : \delta l_3 = 1 : (-3)$$

$$\frac{\delta l_1}{\delta l_2} = \frac{1}{2}$$

$$\delta l_2 = 2 \delta l_1$$

$$\delta U^{(3)} = -\delta U^{(1)} - \delta U^{(2)}$$

$$\delta S = \frac{1}{T^{(1)}} \delta U^{(1)} + \frac{1}{T^{(2)}} \delta U^{(2)} + \frac{1}{T^{(3)}} \delta U^{(3)}$$

$$+ \frac{p^{(1)}}{T^{(1)}} \delta V^{(1)} + \frac{p^{(2)}}{T^{(2)}} \delta V^{(2)} + \frac{p^{(3)}}{T^{(3)}} \delta V^{(3)}$$

$$= \left(\frac{1}{T^{(1)}} - \frac{1}{T^{(3)}} \right) \delta U^{(1)} + \left(\frac{1}{T^{(2)}} - \frac{1}{T^{(3)}} \right) \delta U^{(2)}$$

$$+ \left(\frac{p^{(1)}}{T^{(1)}} + 2 \frac{p^{(2)}}{T^{(2)}} - 3 \frac{p^{(3)}}{T^{(3)}} \right) \delta V^{(1)} = 0$$

$$\rightarrow T^{(1)} = T^{(3)}, \quad T^{(2)} = T^{(3)}$$

$$p^{(1)} + 2 p^{(2)} = 3 p^{(3)}$$

$$1 + 2 \frac{p^{(2)}}{p^{(1)}} = 3 \frac{p^{(3)}}{p^{(1)}}$$

no torque condition

$$p^{(1)} : p^{(2)} : p^{(3)} = ?$$

?

Hamilton's eq. of motion

$$\mathcal{H} = \sum_{j=1}^{3N} \frac{p_j^2}{2m} + V \quad \leftarrow \begin{array}{l} \text{important} \\ \text{to establish} \\ \text{equilibrium} \end{array}$$

$$\dot{p}_j = -\frac{\partial \mathcal{H}}{\partial q_j} = 0$$

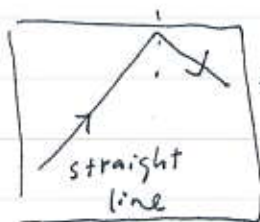
$$p_j = \text{const}$$

$$\dot{q}_j = \frac{\partial \mathcal{H}}{\partial p_j} = \frac{p_j}{m}$$

$$\rightarrow p_j = m \dot{q}_j$$

$$\dot{q}_j = \text{const} = v_j$$

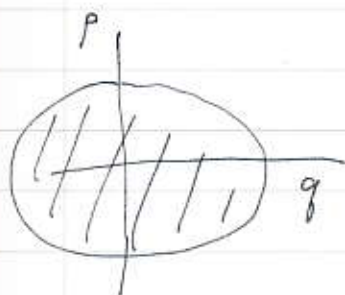
$$q_j = v_j t + \text{const}$$



perfect reflected

how to count the # of states?

$$\Omega(U, V, N) = \frac{1}{N! h^{3N}} \int \dots \int dp_1 dp_2 \dots dp_{3N} dq_1 dq_2 \dots dq_{3N}$$



identical particles solve Gibbs paradox

$$U \leq \mathcal{H} \leq U + \Delta$$

$$\Delta \rightarrow 0$$

$$\oint p dq = nh$$

Bohr-Sommerfeld quantization condition

constant in coordinates $0 \leq q_i \leq L$

$$\Omega = \frac{1}{N! h^{3N}} \int \dots \int dp_1 dp_2 \dots dp_{3N} \cdot V^N$$

$$U \leq \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \dots + \frac{p_{3N}^2}{2m} \leq U + \Delta$$

← volume of a spherical shell in $3N$ dimensions.

$$\Omega \equiv \int \dots \int dp_1 dp_2 \dots dp_{3N}$$

$$2mU \leq p_1^2 + p_2^2 + \dots + p_{3N}^2 \leq 2m(U + \Delta)$$

← the radius of the sphere is from $\sqrt{2mU}$ to $\sqrt{2m(U + \Delta)}$

Euler:

47

$$\Gamma = \int \dots \int dP_1 dP_2 \dots dP_{3N} = R^{3N} \cdot C_{3N}$$

$$P_1^2 + P_2^2 + \dots + P_{3N}^2 \leq 2mU = R^2$$

$$R = \sqrt{2mU}$$

$$C_{3N} = \int_{x_1^2 + x_2^2 + \dots + x_{3N}^2 \leq 1} dx_1 dx_2 \dots dx_{3N} = \frac{\pi^{\frac{3N}{2}}}{\Gamma(\frac{3N}{2} + 1)}$$

$$\Gamma(x+1) = x \Gamma(x)$$

$$\Gamma(x+1) = x!$$

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{1}{2} + 1\right) = \sqrt{\pi}/2$$

$$\left(\frac{3N}{2}\right)!$$

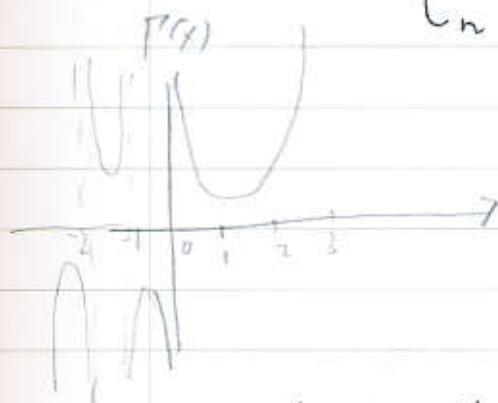
$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$$

volume of n-dimensional sphere is

$$C_n = \frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2} + 1)}$$

n	C _n
1	2
2	π
3	$\frac{4}{3}\pi$



$$\Omega = \frac{1}{h^{3N} N!} V^N \left[\left[2m(U+\Delta) \right]^{\frac{3N}{2}} - 2mU^{\frac{3N}{2}} \right] \frac{\pi^{\frac{3N}{2}}}{\left(\frac{3N}{2}\right)!}$$

Δ small compared to U

$$\approx \frac{1}{h^{3N} N!} V^N \left[\frac{3N}{2} (2mU)^{\frac{3N}{2}-1} 2m\Delta \right] \frac{\pi^{\frac{3N}{2}}}{\left(\frac{3N}{2}\right)!}$$

$$\Delta \sim O(1)$$

$$U \sim O(N)$$

$$S(U, V, N)$$

Take thermodynamic limit

$$N \rightarrow \infty$$

$$V \rightarrow \infty$$

$$U \rightarrow \infty$$

but fixed ratios

$$\frac{V}{N} = \rho = \text{const}$$

$$\frac{U}{N} = u = \text{const}$$

$$\text{is } S(\lambda U, \lambda V, \lambda N) \stackrel{?}{=} \lambda S(U, V, N)$$

No, need

Callan's stronger postulate on
postulate is not
valid for small N !

$$\Omega = \frac{V^N}{h^{3N} N!} \left[(2mU)^{\frac{3N}{2}} \left(\frac{3 \cdot N}{2} \right) \frac{(2m\Delta)}{(2mU)} \right] \frac{\pi^{\frac{3N}{2}}}{\left(\frac{3N}{2} \right) (\frac{3N}{2} - 1)!}$$

$$= \left(\frac{V \cdot (2\pi mU)^{\frac{3}{2}}}{h^3} \right)^N \frac{\Delta}{U} \frac{1}{N! \left(\frac{3N}{2} - 1 \right)!} \rightarrow \text{ignore 1 fact}$$

using Stirling's formula $\ln N! = N \ln N - N + O(\ln N)$

$$S = k_B \ln \Omega = N k_B \ln \frac{V (2\pi mU)^{\frac{3}{2}}}{h^3} + k_B \ln \frac{\Delta}{U} + O(\ln N)$$

$$\left\{ \begin{aligned} & k_B \left(-N \ln N + N - \left(\frac{3N}{2} - 1 \right) \ln \left(\frac{3N}{2} - 1 \right) + \frac{3N}{2} - 1 + O(\ln N) \right) \\ & -N \ln N + N - \frac{3}{2} N \ln \left(\frac{3N}{2} - 1 \right) + \ln \left(\frac{3N}{2} - 1 \right) + \frac{3N}{2} - 1 \dots \\ & \quad - \frac{3}{2} N \left[\ln \left(N - \frac{3}{2} \right) + \ln \frac{3}{2} \right] \\ & \downarrow \\ & -\frac{3}{2} \ln N \left(1 - \frac{3}{N} \right) \\ & k_B \left(-\frac{5}{2} N \ln N + \frac{5}{2} N - \frac{3}{2} N \ln \frac{3}{2} \right) \end{aligned} \right.$$

$$= N k_B \ln \frac{V (2\pi mU)^{\frac{3}{2}}}{h^3} - \frac{5}{2} N k_B \ln N + \frac{5}{2} N k_B - \frac{3}{2} N k_B \ln \frac{3}{2}$$

$$= N k_B \ln \frac{V (2\pi mU)^{\frac{3}{2}}}{N^{\frac{5}{2}} h^3} + \frac{5}{2} N k_B + N k_B \ln \left(\frac{3}{2} \right)^{-\frac{3}{2}}$$

$$= N k_B \ln \left[\frac{V}{N^{\frac{5}{2}} h^3} \left(\frac{4\pi mU}{3} \right)^{\frac{3}{2}} \right] + \frac{5}{2} N k_B$$

Sackur-Tetrode formula (1912)

Otto Sackur, Hugo Tetrode

$$dU = TdS - PdV + \mu dN$$

$$\rightarrow dS = \frac{1}{T} dU + \frac{P}{T} dV - \frac{\mu}{T} dN$$

$$\frac{\partial S}{\partial U} = \frac{1}{T} = \frac{\partial}{\partial U} \left[N k_B \ln U^{\frac{3}{2}} + \text{const} \right] \quad \text{with respect to } U$$

$$= \frac{3}{2} N k_B \frac{1}{U} \rightarrow U = \frac{3}{2} N k_B T$$

$$\frac{\partial S}{\partial V} = \frac{P}{T} = \frac{\partial}{\partial V} [N k_B \ln V + \text{const}] = \frac{N k_B}{V} \rightarrow PV = N k_B T$$

$$\frac{\partial S}{\partial N} = -\frac{\mu}{T} = \frac{\partial}{\partial N} \left[\frac{5}{2} N k_B \ln N + \text{const} \right] = \frac{5}{2} N k_B \frac{1}{N} = \frac{5}{2} k_B \ln N$$

$$\frac{\partial}{\partial N} \left(N k_B \ln \frac{V}{N^{\frac{5}{2}} h^3} \left(\frac{4\pi m U}{3} \right)^{\frac{3}{2}} \right) + \frac{5}{2} N k_B$$

$$= k_B \ln \frac{V}{N^{\frac{5}{2}} h^3} \left(\frac{4\pi m U}{3} \right)^{\frac{3}{2}} + \frac{5}{2} k_B - \frac{5}{2} k_B$$

$$\mu = k_B T \ln \left[\frac{N \lambda^3}{V} \right]$$

$$\lambda = \frac{h N^{\frac{1}{2}}}{\left(\frac{4\pi m U}{3} \right)^{\frac{1}{2}}}$$

$$= \frac{h N^{\frac{1}{2}}}{\left(\frac{4\pi m}{3} \frac{3}{2} N k_B T \right)^{\frac{1}{2}}}$$

$$\lambda = \frac{h}{\sqrt{2\pi m k_B T}} \leftarrow \text{thermal wavelength}$$

Does S satisfies 3rd law of thermodynamics? why?

Virial Theorem
Equipartition Theorem

The degree of freedom
is S .

$$k_B T = \langle \vec{u} \cdot \vec{\nabla} H \rangle$$

microcanonical ensemble

$$\vec{u} = (u_1, u_2, \dots, u_{2S})$$

$$\text{such that } \vec{\nabla} \cdot \vec{u} = 1$$

$$\vec{\nabla} = \left(\frac{\partial}{\partial q_1}, \frac{\partial}{\partial q_2}, \dots, \frac{\partial}{\partial q_S}, \frac{\partial}{\partial p_1}, \frac{\partial}{\partial p_2}, \dots, \frac{\partial}{\partial p_S} \right)$$

E.g. Let $H = \sum_{j=1}^S \frac{p_j^2}{2m_j} + V(q_1, \dots, q_S)$

$$\vec{u} = (0, 0, \dots, 0, 0, \frac{p_j}{m_j}, 0, \dots, 0)$$

$$\vec{\nabla} \cdot \vec{u} = 1$$

$$\vec{\nabla} H = \left(\frac{\partial H}{\partial q_1}, \frac{\partial H}{\partial q_2}, \dots, \frac{\partial H}{\partial q_S}, \frac{p_1}{m_1}, \frac{p_2}{m_2}, \dots, \frac{p_S}{m_S} \right)$$

$$k_B T = \left\langle \frac{p_j^2}{m_j} \right\rangle$$

$\frac{p_j}{m_j}$ 1st
q 2nd

$$\left\langle \frac{1}{2} m_j \dot{q}_j^2 \right\rangle = \frac{1}{2} k_B T$$

average kinetic energy of
each degree of freedom is
 $\frac{1}{2} k_B T$

$$P = \begin{pmatrix} p \\ q \end{pmatrix} \quad \vec{\nabla} = \begin{pmatrix} \frac{\partial}{\partial p} \\ \frac{\partial}{\partial q} \end{pmatrix}$$

configurational temperature

$$\vec{u} = (0, \dots, q_j, 0, 0, \dots, 0)$$

$$k_B T = \left\langle q_j \frac{\partial H}{\partial q_j} \right\rangle = - \left\langle q_j F_j \right\rangle \quad F_j = - \frac{\partial V}{\partial q_j}$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V, N} \quad \leftarrow \text{definition of temperature}$$

Boltzmann's principle

Date

$$S(U, V, N) \equiv k_B \ln \Omega(U, V, N)$$

$$\Omega(U, V, N) = \frac{1}{N! h^{3N}} \int_{U < \mathcal{H} < U + \delta} dq_1 \dots dq_{3N} dp_1 \dots dp_{3N}$$

$$\text{Let } \Gamma(E) = \int_{\mathcal{H} < E} dq \dots dq$$

$$\frac{1}{T} = \frac{\partial}{\partial U} k_B \ln \Omega = k_B \frac{1}{\Omega} \frac{\partial \Omega}{\partial U}$$

$$\frac{\partial \Omega}{\partial U} = \frac{\Omega(U + \Delta U) - \Omega(U)}{\Delta U} \quad \Delta U \rightarrow 0$$

$$= \frac{\Gamma(U + \Delta U + \delta) - \Gamma(U + \Delta U) - \Gamma(U + \delta) + \Gamma(U)}{\Delta U}$$

$$= \Gamma(U + \delta)$$

$$\frac{\Gamma(U + \delta + \Delta U) - \Gamma(U + \delta)}{\Delta U} = \frac{\Gamma(U + \delta) - \Gamma(U)}{\Delta U}$$

$$S(U, V, N) = k_B \ln \left(\Gamma(U + \delta, V, N) - \Gamma(U, V, N) \right)$$

Alternative expression

(same as use Ω in the thermodynam limit)

$$S(U, V, N) = k_B \ln \Gamma(U, V, N) \quad \text{diff from } \ln \Omega \text{ by at most } \log N!$$

$$\frac{1}{T} = \frac{k_B}{\Gamma} \frac{\partial \Gamma}{\partial U} = \frac{k_B}{\Gamma} \frac{\Gamma(U + \Delta U, V, N) - \Gamma(U, V, N)}{\Delta U}$$

$$\Delta U \rightarrow 0$$

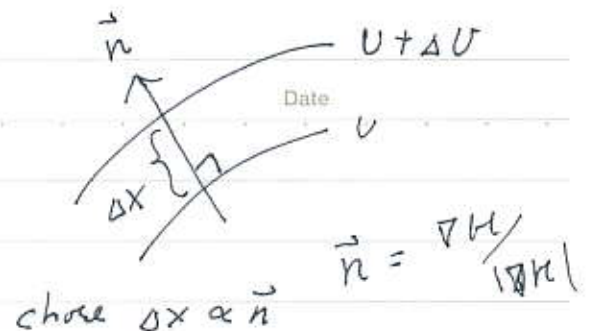
$$\omega = \frac{\Omega}{\Delta U} = \frac{\Delta \Gamma}{\Delta U} = \frac{1}{\Delta U} \int d\Gamma \leftarrow \text{volume}$$

$U < H < U + \Delta U$

take $\Delta U \rightarrow 0$
limit

$$= \int_{H=U} d\sigma \cdot \frac{1}{|\nabla H|}$$

$H=U$
surface



$$\Delta U = \nabla H \cdot \Delta \mathbf{x} = |\nabla H| \cdot \Delta x$$

$$d\Gamma = d\sigma \cdot \Delta x$$

$$\frac{1}{k_B T} = \frac{1}{\Gamma} \int_{H=U} \frac{d\sigma}{|\nabla H|}$$

proper measure
on the surface
of $H=U$

$$= \frac{d\sigma \cdot \Delta U}{|\nabla H|}$$

$$= \frac{1}{\int d\Gamma} \int_{H < U} \frac{d\sigma}{|\nabla H|}$$

$H < U$

$$= \frac{\int \frac{d\sigma}{|\nabla H|}}{\int \nabla \cdot \vec{u} d\Gamma} = \frac{\int \frac{d\sigma}{|\nabla H|}}{\int \vec{u} \cdot \vec{n} d\sigma}$$

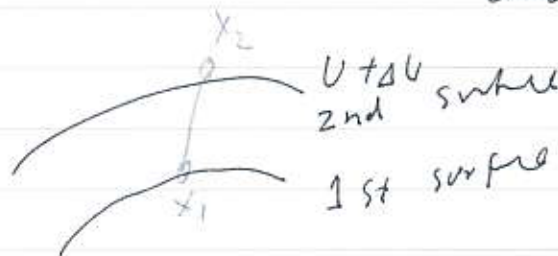
Gauss theorem in 3D divergence theorem

$$k_B T = \frac{1}{\int \frac{d\sigma}{|\nabla H|}} \int \vec{u} \cdot \nabla H \frac{d\sigma}{|\nabla H|} = \langle \vec{u} \cdot \vec{\nabla H} \rangle$$

microcanonical average

$$U_2 = U + \Delta U = H(\Gamma_2)$$

$$U_1 = U = H(\Gamma_1)$$



$$\Delta U = H(\Gamma_2) - H(\Gamma_1)$$

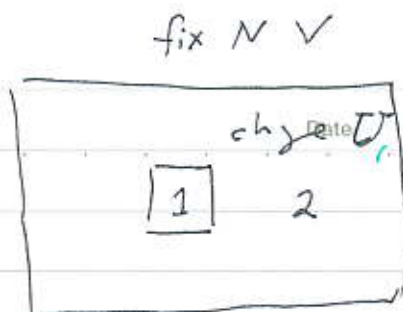
$$= \sum_i \frac{\partial H}{\partial q_i} dq_i + \sum_j \frac{\partial H}{\partial p_j} dp_j = \left. \nabla H \right|_{\Gamma_1} \cdot \Delta \mathbf{x}$$

if Γ_1 & Γ_2 on the same surface $\Delta U = 0$ so $\vec{n} \cdot \Delta \mathbf{x}$

$$\Delta \mathbf{x} = (dq_1, dq_2, \dots, dp_1, \dots, dp_n)$$

Canonical ensemble

phase space pt Γ



$U_1 + U_2 = U$
free

$$H_1(q_1, q_2 \dots p_1, p_2 \dots) \quad \Gamma_1 = (q_1, q_2 \dots)$$

$$H_2(q'_1, q'_2 \dots) \quad \Gamma_2 = (q'_1, q'_2 \dots)$$

$$H(q_1, q_2 \dots q'_1, q'_2 \dots)$$

$$H = H_1 + H_2 + V$$

$\leftarrow$ small $\frac{2}{3}$

$O(N)$ $O(N)$ $O(N)$

finite $\rightarrow$ too

start from microscopic Microcanonical ensemble

$$\rho(\Gamma_1, \Gamma_2) = \begin{cases} \text{const} & U < H < U + \Delta \\ 0 & \end{cases}$$

$\rho(\Gamma_1, \Gamma_2) d\Gamma_1 d\Gamma_2$ gives the probab. that state is in side volume $d\Gamma_1 d\Gamma_2$ and $\Gamma = (\Gamma_1, \Gamma_2)$

$\uparrow$ joint distribution

Take $N_2 \rightarrow \infty$ (also $N_1 \rightarrow \infty$)

what is probability that system 1 is at Γ_1 in volume $d\Gamma_1$ regardless system 2? $\rho \approx \text{const}$

$$\rho_1(\Gamma_1) d\Gamma_1 = d\Gamma_1 \int \rho(\Gamma_1, \Gamma_2) d\Gamma_2 \propto \Omega_2(E - E_1)$$

$\rightarrow$ cannot E else

integr on Γ_2

$$\rho(\Gamma_1, \Gamma_2) \approx \rho_1(E_1) \rho_2(\Gamma_2, E_2) \quad U < E_1 + E_2 = U + \Delta$$

$\uparrow$
micro
call
 E free

$\downarrow$

$$U - E_1 < E_2 < U - E_1 + \Delta$$

$$P_1(\Gamma_1) \propto \Omega_2(E - E_1)$$

the probability of the system that can exchange energy with a heat bath is proportional to the number of micro states of the bath system at a energy of $E_2 = E - E_1$

$$k_B \ln \Omega_2(E - E_1) = S_2(E - E_1)$$

$$= S_2(E) - E_1 \frac{\partial S_2(E)}{\partial E} - E_1^2 \frac{\partial^2 S_2(E)}{\partial E^2} + \dots$$

$$= S_2(E) - E_1/T + \text{small?}$$

$\frac{\partial^2 S_2}{\partial E^2} \propto O(1/N)$

$$\ln \Omega_2 = \frac{S_2}{k_B} - \frac{E_1}{k_B T}$$

↑
unit

$$\frac{1}{k_B T} = \beta$$

$$\Omega_2 = e^{\frac{S_2}{k_B} - \frac{E_1}{k_B T}} \propto e^{-\beta E_1}$$

$$k_B \ln \Omega = S(U)$$

$$\rightarrow P_1(\Gamma_1) \propto e^{-\beta E_1(\Gamma_1)}$$



Relation between canonical ensemble & thermodynamics

def. $Z = \int d\Gamma e^{-\beta H(\Gamma)} = \sum_i e^{-\beta E_i} \int d\Gamma = \sum_i e^{-\beta E_i} \Omega(E_i)$

$\int dE \Omega(E) e^{-\beta E}$

$$Q = Z = \sum_i e^{-\beta E_i} + S(E_i)/k_B$$

$$\Delta E^2 = k_B T^2 C_V$$

$U \sim O(N)$
 $S \sim O(N)$

$$N \rightarrow \infty \quad \ln Z = -\min (\beta U + S(U)/k_B)$$

$$\ln Z = -\min_E (\beta E - S(E)/k_B)$$

let the min be $U = \langle E \rangle$

$$F = -k_B T \ln Z = U - TS$$

$F = U - TS$

Helmholtz free energy

Do this better!

Quantum statistical mechanics

(see chap 8, K. Huang, 2nd ed)

Date

microcanonical

$$\rho \propto \sum |\Phi_n\rangle \langle \Phi_n|$$

$$U < E_n < U + \Delta$$

where

$$\hat{H} |\Phi_n\rangle = E_n |\Phi_n\rangle$$

eigen states of energy

canonical

$$\rho \propto e^{-\beta \hat{H}}$$

grand canonical

$$\rho \propto e^{-\beta (\hat{H}_N - \mu \hat{N})}$$

↑ operator
denote # of
particleseigenvalues of $\hat{N}$ is
0, 1, 2, 3, ...genl properties of
density operator ρ

$$(1) \rho \text{ is positive definite } (\hat{\rho} |\psi\rangle = w |\psi\rangle)$$

$$(2) \text{Tr}(\rho) = 1 \quad (\text{proportionality const } w \geq 0 \text{ is fixed by normalization})$$

$$(3) \langle A \rangle = \text{Tr}(\rho A) = \sum_j \langle \psi_j | \rho A | \psi_j \rangle$$

↑ some complete set
in Hilbert space / Fock space

classical

$$\int \frac{d\Gamma}{N! h^{3N}}$$

$$Z = \int \frac{d\Gamma}{N! h^{3N}} e^{-\beta H(\Gamma)}$$

quantum

$$\text{Tr}(\dots)$$

$$Z = \text{Tr}(e^{-\beta \hat{H}})$$

Three most common useful ensembles

microcanonical

$$p(\Gamma) \propto \begin{cases} \text{const, } U < H(\Gamma) < U + \Delta \\ 0 \end{cases}$$

entropy

$$S(U, V, N) = k_B \ln \Omega \uparrow$$

of states / phase space volume

canonical

$$p(\Gamma) = e^{-\beta H(\Gamma)} / Z$$

partition function

$$Z = \int \frac{d\Gamma}{N! h^{3N}}$$

$$F(T, V, N) = -k_B T \ln Z$$

$\longleftrightarrow$

$$F = U - TS$$

Helmholtz free energy

$$dU = TdS - pdV + \mu dN$$

Gibbs free energy

$$G = G(T, p, N) = U - TS + pV$$

$$dG = TdS - pdV + \mu dN - TdS - SdT - \mu dN - Nd\mu = -SdT - pdV - Nd\mu$$

$$\rightarrow -S = - \left(\frac{\partial G}{\partial T} \right)_{p, N}$$

$$-p = - \left(\frac{\partial G}{\partial V} \right)_{T, N}$$

$$\mu = \left(\frac{\partial G}{\partial N} \right)_{T, p}$$

grand-canonical

$$p(\Gamma, N) \propto e^{-\beta(H - \mu N)}$$

grand partition function

$$\Xi = \sum_{N=0,1,2,\dots} \int \frac{d\Gamma}{N! h^{3N}} e^{-\beta(H - \mu N)} = \sum_{N=0,1,2,\dots} Z_N e^{\beta \mu N}$$

grand potential

$$U - TS - \mu N = \Phi = \Psi$$

$$= -k_B T \ln \Xi(T, V, \mu)$$

grand potential notation

by Reichl

by Callen

by Plischke

Date

average # of particles because the N fluctuating in grand canonical ensemble

$$\langle N \rangle = - \left(\frac{\partial \Phi}{\partial \mu} \right)_{T, V}$$

Applications

harmonic

System of coupled oscillators (i.e. thermal vibrations of solids)

$$H = \sum_{j=1}^N \frac{p_j^2}{2m_j} + \frac{1}{2} \sum_{i,j=1}^N q_i K_{ij} q_j$$

lattice spring constant

eg. monatomic...

column vectors

$$p_j = m_j \dot{q}_j$$

$$\mathbf{q} = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_s \end{pmatrix}$$

$$K = \begin{pmatrix} 2K & -K & 0 \\ -K & 2K & -K \\ 0 & -K & 2K \end{pmatrix}$$

$$K^T = K$$

Transform into normal-mode representation

Jacobian for p, q to $u, \dot{u}$

Let

$$u_j = \sqrt{m_j} q_j$$

$$\begin{vmatrix} \frac{\partial u}{\partial q} & \frac{\partial u}{\partial p} \\ \frac{\partial \dot{u}}{\partial q} & \frac{\partial \dot{u}}{\partial p} \end{vmatrix} = \begin{vmatrix} \sqrt{m} & 0 \\ 0 & \frac{1}{\sqrt{m}} \end{vmatrix} = 1$$

$$\sum \frac{1}{2} m_j \dot{q}_j^2 \rightarrow \sum \frac{1}{2} \dot{u}_j^2$$

$$\sum q_i K_{ij} q_j \rightarrow \sum u_i \frac{K_{ij}}{\sqrt{m_i m_j}} u_j = \sum u_i \bar{K}_{ij} u_j$$

det(A) = 1

so du, dv = \frac{\partial u}{\partial q} dq + \frac{\partial u}{\partial p} dp

$$H = \frac{1}{2} \dot{u}^T \dot{u} + \frac{1}{2} u^T \bar{K} u$$

matrix (orthogonal)

$$\text{Let } A Q = u$$

$$A A^T = I$$

$$\det(A) = 1$$

eigenvalue problem

$$H = \frac{1}{2} \dot{Q}^T A^T A \dot{Q} + \frac{1}{2} Q^T A^T \bar{K} A Q$$

$$= \frac{1}{2} \dot{Q}^T \dot{Q} + \frac{1}{2} Q^T \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \\ \vdots & \vdots \end{bmatrix} Q$$

$$A^T \bar{K} A = \begin{pmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \\ \vdots & \vdots \end{pmatrix}$$

A diagonal

$$= \sum_j \frac{1}{2} (\dot{Q}_j^2 + \omega_j^2 Q_j^2)$$

Chap 6 of Goldstein 3rd ed.

 $\bar{K}$

compute in canonical ensemble (classical version)

$$Z = \frac{1}{h^N} \int d\dot{q}_1 d\dot{q}_2 \dots d\dot{q}_N \int dq_1 dq_2 \dots dq_N e^{-\beta \sum_{j=1}^N \left(\frac{1}{2} \dot{q}_j^2 + \frac{1}{2} \omega_j^2 q_j^2 \right)}$$

why not $N!$

$$= \frac{1}{h^N} \left[\int d\dot{q}_1 e^{-\frac{\beta}{2} \dot{q}_1^2} \right]^N \int dq_1 e^{-\frac{\beta \omega_1^2}{2} q_1^2} \int dq_2 e^{-\frac{\beta \omega_2^2}{2} q_2^2} \dots$$

$$= \frac{1}{h^N} \left(\frac{1}{\sqrt{\beta}} \int_{-\infty}^{\infty} dy e^{-\frac{y^2}{2}} \right)^N \prod_{j=1}^N \left(\sqrt{\frac{2\pi}{\beta \omega_j^2}} \right) = \prod_{j=1}^N \left(\frac{k_B T}{h \omega_j} \right)$$

$$\beta \dot{q}_1^2 = y^2$$

$$\sqrt{\beta} \dot{q}_1 = y$$

$$= (2\pi)^N \frac{1}{\beta^N} \prod_{j=1}^N \frac{1}{\omega_j h} = \left(\frac{2\pi}{\beta} \right)^N \prod_{j=1}^N \frac{1}{\omega_j} \left(\frac{1}{2\pi h} \right)$$

$$F = -\frac{1}{\beta} \quad F(T, N) = -\frac{1}{\beta} \left[\ln \left(\frac{(2\pi)^N}{\beta^N} \right) \oplus \sum_{j=1}^N \ln \omega_j \right]$$

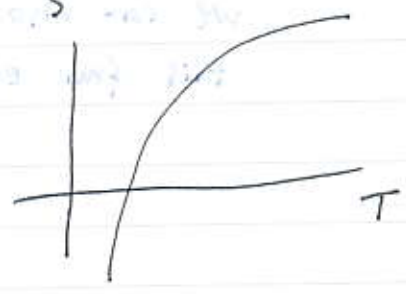
see back put

$$S = -\frac{\partial F}{\partial T} = + \frac{\partial}{\partial T} \left[-k_B T \left(N \ln \frac{1}{2\pi h} + \sum_{j=1}^N \ln \omega_j \right) \right]$$

$$= \frac{\partial}{\partial T} \left[k_B T \left(N \ln \left(\frac{k_B T}{2\pi h} \right) + \sum_{j=1}^N \ln \omega_j \right) \right]$$

$$= k_B \left[N \ln \left(\frac{k_B T}{2\pi h} \right) + \sum_{j=1}^N \ln \omega_j \right] + k_B N$$

$$S = + N k_B \ln T + \text{const.}$$



$$U = F + TS$$

$$= -k_B T \left[N \ln (2\pi k_B T) + \sum_{j=1}^N \ln \omega_j \right] + k_B T \left[N \ln (2\pi k_B T) + \sum_{j=1}^N \ln \omega_j \right]$$

$$= N k_B T + N k_B T$$

$$Z = \prod_{j=1}^N \frac{k_B T}{h \omega_j}$$

$$F = -k_B T \ln Z$$

$$= -k_B T \sum_{j=1}^N \ln \left(\frac{k_B T}{h \omega_j} \right)$$

$$S = - \frac{\partial F}{\partial T} = k_B \sum_{j=1}^N \ln \left(\frac{k_B T}{h \omega_j} \right) + k_B N$$

$$= k_B \sum_{j=1}^N \ln \left(\frac{k_B T e}{h \omega_j} \right)$$

$$U = F + TS$$

$$= -k_B T \ln Z + T \left(\frac{\partial}{\partial T} k_B T \ln Z \right)$$

$$= -k_B T \ln Z + k_B T \ln Z + k_B T^2 \frac{\partial \ln Z}{\partial T}$$

$$= k_B T^2 \frac{\partial \ln Z}{\partial T} = - \frac{\partial \ln Z}{\partial \beta}$$

$$= k_B T^2 \left[\frac{1}{T} \cdot N \right] = N k_B T$$

$$= - \frac{\partial T}{\partial \left(\frac{1}{k_B T} \right)}$$

we can also get
this from equipartition theorem

$$= k_B \frac{\partial T}{\frac{1}{T^2}} = k_B T^2 \frac{\partial}{\partial T} \ln Z$$

heat capacity

$$C_V = \frac{dU}{dT} = Nk_B = nR \leftarrow \text{indep of } T \text{ for solids}$$

Date

Quantum version

$$Z = \text{Tr } e^{-\beta \hat{H}} = \sum_K \langle \phi_K | e^{-\beta \hat{H}} | \phi_K \rangle$$

 $\phi_K(Q_1, Q_2 \dots Q_N) \leftarrow \text{some complete set of wave functions}$

Use energy eigen states

$$\hat{H} \phi_K(Q_1, Q_2 \dots Q_N) = E_K \phi_K(Q_1, Q_2 \dots Q_N)$$

$$\begin{aligned} \hat{H} &= \sum_j \frac{1}{2} (\dot{Q}_j^2 + \omega_j^2 Q_j^2) \\ &= \sum_j \hat{h}_j(Q_j) \end{aligned}$$

$$\dot{Q}_j \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial Q_j}$$

$$[Q_j, \dot{Q}_\ell] = i\hbar \delta_{j\ell}$$

$$\hat{h}_j(Q_j) = -\frac{1}{2}\hbar^2 \frac{\partial^2}{\partial Q_j^2} + \frac{1}{2}\omega_j^2 Q_j^2$$

$$Q = \sqrt{m} x$$

$$\hat{h}_j \phi_n = \hbar \omega_j (n + \frac{1}{2})$$

harmonic oscillator

$$\text{Let } \phi_K(Q_1, Q_2 \dots Q_N) = \prod_j \phi_{n_j}(Q_j)$$

$$\text{quantum number } K = (n_1, n_2, n_3 \dots n_N)$$

$$n_j = 0, 1, 2, 3, \dots$$

$$E_K = \sum_{j=1}^N \hbar \omega_j (n_j + \frac{1}{2})$$

$$Z = \sum_{(n_1, n_2, \dots, n_N)} e^{-\beta \sum_j \hbar \omega_j (n_j + \frac{1}{2})}$$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad 73$$

$$= \prod_j \sum_{n_j=0}^{\infty} e^{-\beta \hbar \omega_j (n_j + \frac{1}{2})} = \prod_j e^{-\beta \frac{\hbar \omega_j}{2}} \frac{1}{1 - e^{-\beta \hbar \omega_j}}$$

$$F = -k_B T \ln Z$$

$$= -k_B T \left\{ \sum_j \left[\left(-\beta \frac{\hbar \omega_j}{2} \right) - \ln (1 - e^{-\beta \hbar \omega_j}) \right] \right\}$$

Another approx

$$Z = \text{Tr} e^{-\beta \sum_j \hbar \omega_j (\hat{a}_j^\dagger \hat{a}_j + \frac{1}{2})}$$

$$\hat{a}_j^\dagger \hat{a}_j |n_j\rangle = n_j |n_j\rangle$$

$$H = \sum_j \hbar \omega_j (\hat{a}_j^\dagger \hat{a}_j + \frac{1}{2})$$

$$\hat{a}_j = \hat{Q}_j$$

$$\hat{a}_j = \sqrt{\frac{m}{2\hbar}} \left(\hat{Q}_j + \frac{i}{\omega_j} \hat{P}_j \right)$$

$$[\hat{a}_j, \hat{a}_{j'}^\dagger] = \delta_{jj'}$$

$$A(t) = e^{\mathcal{L}t} A = A + t\mathcal{L}A + \frac{1}{2}t^2\mathcal{L}^2A + \dots$$

$$\dot{A}(t) = \mathcal{L} e^{\mathcal{L}t} A = \mathcal{L} A(t) = [A(t), H]$$

Date

$$L f = \left[f, \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2 \right]$$

$$= [f, p^2] \frac{1}{2m} + [f, q^2] \frac{1}{2} m \omega^2$$

$$= p [f, p] \frac{1}{2m} + [f, p] p \frac{1}{2m}$$

$$+ q [f, q] \frac{1}{2} m \omega^2 + [f, q] q \frac{1}{2} m \omega^2$$

Let $f = q$

$$L q = \frac{1}{m} p$$

$$L^2 q = \frac{1}{m} L p = -\omega^2 q$$

$$L^3 q = L(-\omega^2) q = (-\omega^2) \frac{1}{m} p$$

$$L^4 q = (-\omega^2)^2 q$$

$$q(t) = q + \frac{t}{m} p + \frac{1}{2!} t^2 (-\omega^2) q + \frac{1}{3!} t^3 (-\omega^2) \frac{p}{m} + \frac{1}{4!} t^4 (-\omega^2)^2 q + \dots$$

$$= \left[1 - \frac{1}{2!} (\omega t)^2 + \frac{1}{4!} (\omega t)^4 + \dots \right] q$$

$$+ \left[t + \frac{1}{3!} (-\omega t)^3 + \dots \right] \frac{p}{m}$$

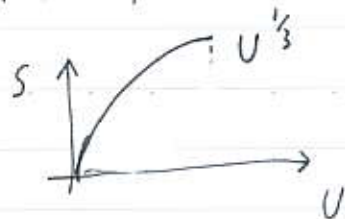
$$= q \cos \omega t + \frac{p}{m \omega} \sin \omega t + ?$$

- P III { ① homogen 4 order 1 ✓
 ② monotonic increase with U ✓

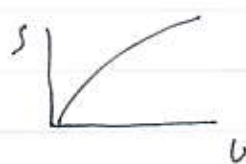
Assignment 1

P IV { ③ $\frac{1}{T} = \frac{\partial S}{\partial U} = U^{-\frac{2}{3}} \quad T=0 \quad S=0 \quad \checkmark$
 ④ concave $\frac{\partial^2 S}{\partial U^2} < 0$

a) ✓



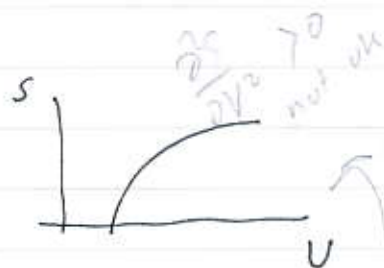
b) X



$$\left(\frac{NU}{V}\right)^{\frac{2}{3}} \rightarrow \left(\frac{\lambda^2}{\lambda}\right)^{\frac{2}{3}} = \lambda^{\frac{2}{3}}$$

Not homogen

c) ✓



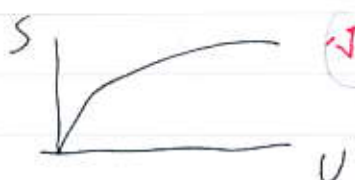
$$\frac{\partial S}{\partial U} = \frac{1}{(NU + bV^2)} = \infty \quad S=0 \quad \checkmark$$

d) X



P III violated not increase with U

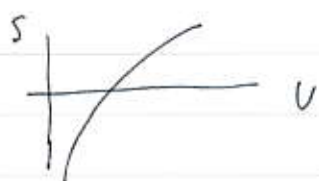
e) ✓



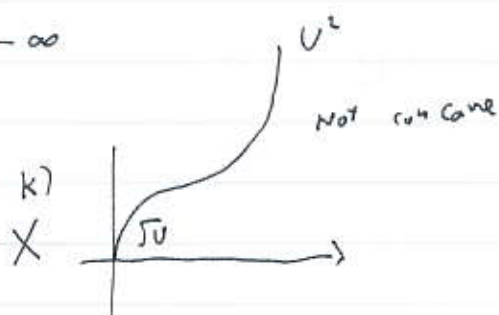
✓ depends
not concave

$$\frac{\partial S}{\partial U} = U^{-\frac{3}{5}}$$

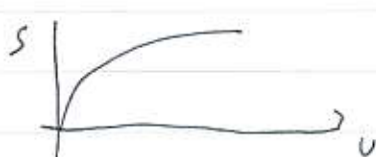
f) X



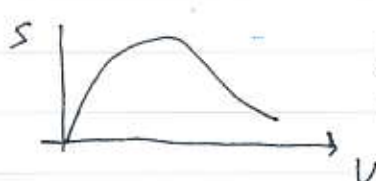
S → -∞



g) ✓

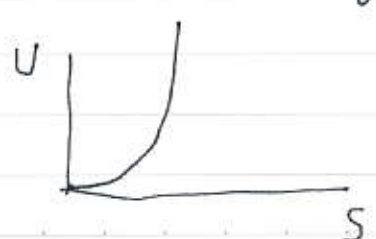


h) X

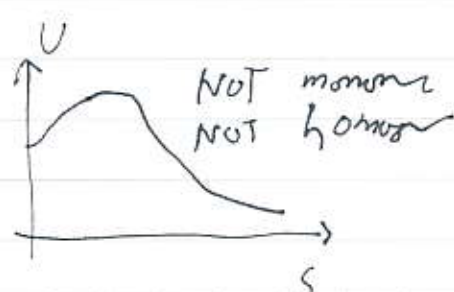


NOT monotonic
NOT homogen

i) ✓



j) X



NOT monotonic
NOT homogen